

ECE 161C: Introduction to Image Processing Image Reconstruction

Rahul Parhi
Assistant Professor, ECE, UCSD

Spring 2026

Outline

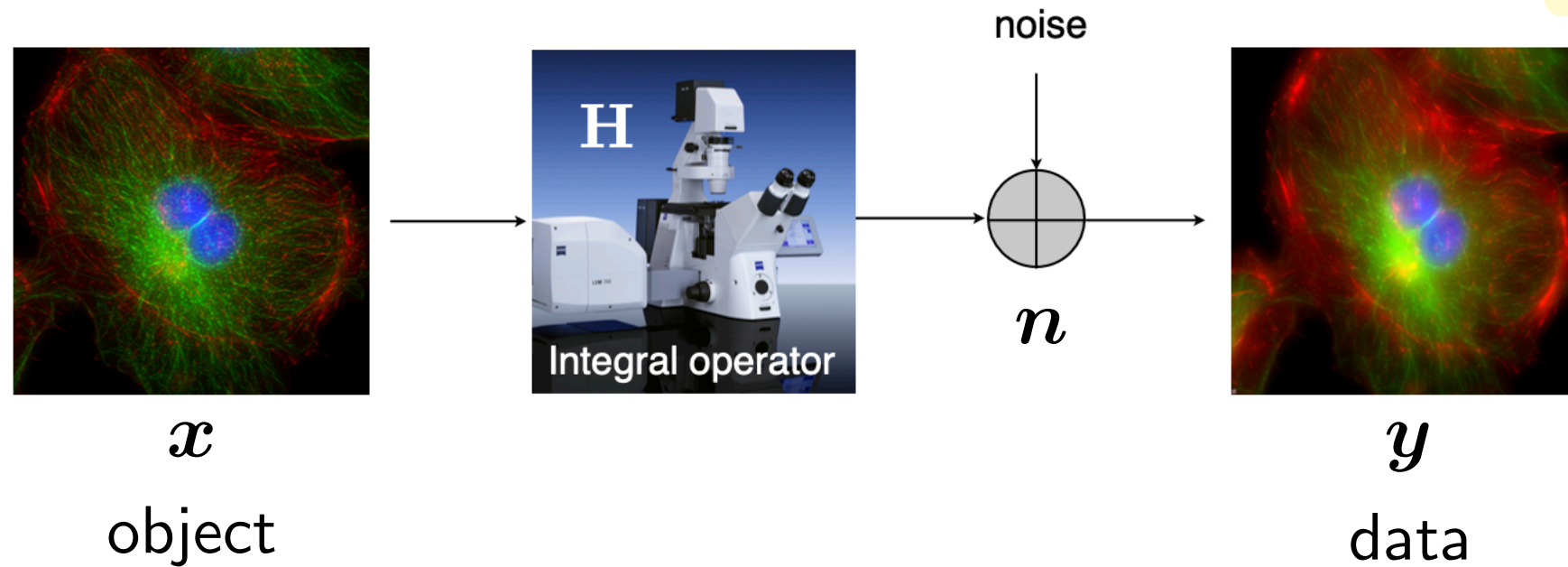
- Image reconstruction as an inverse problem
- Imaging operators and modalities
- Discretization of inverse problems



Image Reconstruction as an Inverse Problem

- Linear forward model

$$y = \mathbf{H}x + n$$



Given y , reconstruct x

“invert $\mathbf{H}$ ”

Why is this problem hard?

- noise amplification
- difficult to invert $\mathbf{H}$ (large or non-square)
- all interesting inverse problems are **ill-posed**

- Backprojection (poor man's solution): $\hat{x} = \mathbf{H}^T y$

What are some examples of inverse problems?

Imaging Operators and Modalities

- Fourier Transform
- Windowing
- Convolution
- Radon Transform

Mathematical Formulation of Forward Models

- Unknown object: $s(\mathbf{x})$, $\mathbf{x} \in \mathbb{R}^2$ (or more generally $\mathbf{x} \in \mathbb{R}^d$)
defined in the analog domain

$$s \in L^2(\mathbb{R}^2) \quad (\text{finite-energy objects})$$

- Imaging operator/forward model: $H\{s\} = \mathbf{y} \in \mathbb{R}^M$
from analog to discrete

$$H : L^2(\mathbb{R}^2) \rightarrow \mathbb{R}^M$$

- Boundedness (BIBO) and linearity assumption:

$$\|H\{s\}\|_2 \leq C\|s\|_{L^2}$$

$$H\{\alpha_1 s_1 + \alpha_2 s_2\} = \alpha_1 H\{s_1\} + \alpha_2 H\{s_2\}, \text{ for all } s_1, s_2 \in L^2(\mathbb{R}^2), \alpha_1, \alpha_2 \in \mathbb{R}$$

$$\Rightarrow y_m = \langle \eta_m, s \rangle = \int_{\mathbb{R}^2} \eta_m(\mathbf{x}) s(\mathbf{x}) \, d\mathbf{x} \quad (\text{Riesz representation theorem})$$

$\{\eta_m\}_{m=1}^M$ are called the **analysis functions**

Basic Operator: Fourier Transform

$$\mathcal{F} : L^2(\mathbb{R}^2) \rightarrow L^2(\mathbb{R}^2)$$

$$\hat{s}(\boldsymbol{\omega}) = \mathcal{F}\{s\}(\boldsymbol{\omega}) = \int_{\mathbb{R}^2} s(\boldsymbol{x}) e^{-j\boldsymbol{\omega}^\top \boldsymbol{x}} d\boldsymbol{x}$$

Reconstruction formula (inverse Fourier transform)

$$s(\boldsymbol{x}) = \mathcal{F}^{-1}\{\hat{s}\}(\boldsymbol{x}) = \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} \hat{s}(\boldsymbol{\omega}) e^{j\boldsymbol{\omega}^\top \boldsymbol{x}} d\boldsymbol{\omega}$$

Exercise: What would be the analysis functions of the forward model that samples the Fourier transform at $\{\boldsymbol{\omega}_m\}_{m=1}^M$?

Basic Operator: Windowing

$$W : L^2(\mathbb{R}^2) \rightarrow L^2(\mathbb{R}^2)$$

$$W\{s\}(\boldsymbol{x}) = w(\boldsymbol{x})s(\boldsymbol{x})$$

w is a positive and bounded window function:

$$w(\boldsymbol{x}) \geq 0 \text{ and } w(\boldsymbol{x}) \leq C \text{ for all } \boldsymbol{x} \in \mathbb{R}^2$$

- Special case: Modulation

$$w(\boldsymbol{x}) = e^{j\boldsymbol{\omega}_0^T \boldsymbol{x}}$$

Structured Illumination Microscopy (SIM)

Magnetic Resonance Imaging

- Simplified forward model for MRI

$$\hat{s}(\boldsymbol{\omega}_m) = \int_{\mathbb{R}^2} s(\boldsymbol{x}) e^{-j\boldsymbol{\omega}_m^T \boldsymbol{x}} d\boldsymbol{x}$$

(sampling of the Fourier transform)



- More-realistic forward model for MRI

$$\hat{s}_w(\boldsymbol{\omega}_m) = \int_{\mathbb{R}^2} w(\boldsymbol{x}) s(\boldsymbol{x}) e^{-j\boldsymbol{\omega}_m^T \boldsymbol{x}} d\boldsymbol{x}$$

(sampling of the short-time (short-space?) Fourier transform)

(window models the effect of the coil)

Basic Operator: Convolution

$$\mathbf{H} : L^2(\mathbb{R}^2) \rightarrow L^2(\mathbb{R}^2)$$

$$\mathbf{H}\{s\}(\boldsymbol{x}) = (h * s)(\boldsymbol{x}) = \int_{\mathbb{R}^2} h(\boldsymbol{x} - \boldsymbol{y}) s(\boldsymbol{y}) \, \mathrm{d}\boldsymbol{y}$$

Impulse response: $h = \mathbf{H}\{\delta\}$

- Convolution as frequency-domain product: $(h * s)(\boldsymbol{x}) \xleftrightarrow{\mathcal{F}} \hat{h}(\boldsymbol{\omega}) \hat{s}(\boldsymbol{\omega})$

Exercise: What would be the analysis functions of the forward model that samples the convolution at $\{\boldsymbol{x}_m\}_{m=1}^M$?

Basic Operator: Radon Transform

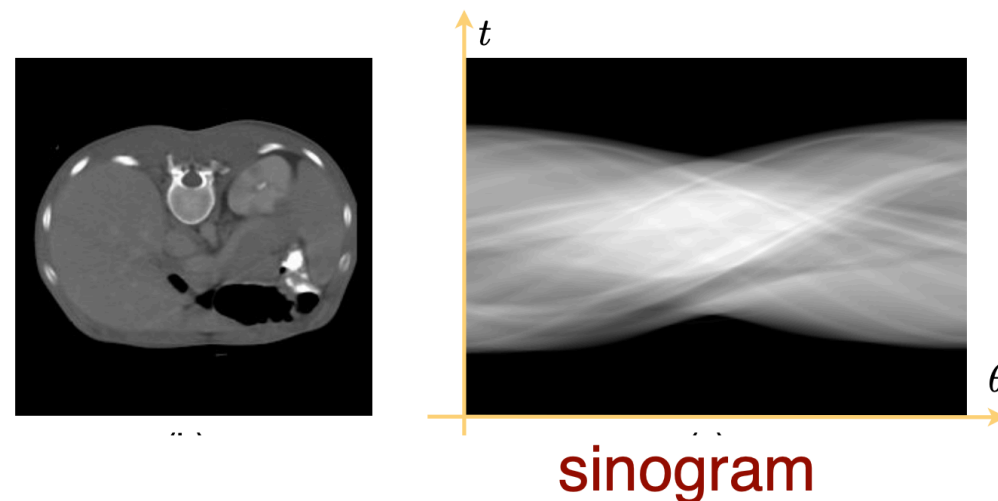
- Radon transform (line integrals)

Notation: $\boldsymbol{\theta} = (\cos \theta, \sin \theta) \in \mathbb{R}^2$

A line in $\mathbb{R}^2$ can be represented by all $\boldsymbol{x} \in \mathbb{R}^2$ such that

$$\boldsymbol{\theta}^\top \boldsymbol{x} = t \quad \Leftrightarrow \quad x \cos \theta + y \sin \theta = t$$

$$\mathcal{R}_\theta\{f\}(t) = \int_{\mathbb{R}^2} f(\boldsymbol{x}) \delta(\boldsymbol{\theta}^\top \boldsymbol{x} - t) \, d\boldsymbol{x}$$



Exercise: What would be the analysis functions of the forward model that samples the Radon transform at $\{(\theta_m, t_m)\}_{m=1}^M$?

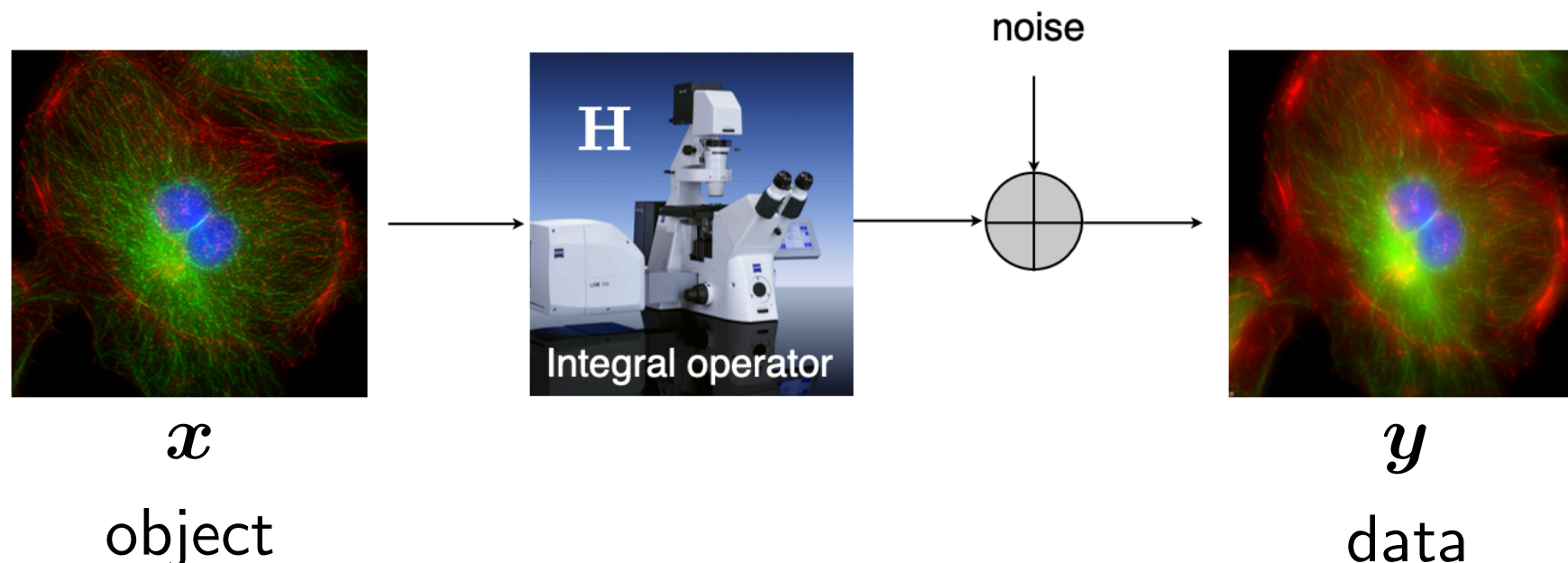
Panorama of Imaging Modalities

Modality	Radiation	Forward model	Variations
2D or 3D tomography	coherent x-ray	$y_i = R_{\theta_i} x$	parallel, cone beam, spiral sampling
3D deconvolution microscopy	fluorescence	$y = Hx$	brightfield, confocal, light sheet
structured illumination microscopy (SIM)	fluorescence	$y_i = HW_i x$ H: PSF of microscope W_i : illumination pattern	full 3D reconstruction, non-sinusoidal patterns
Positron Emission Tomography (PET)	gamma rays	$y_i = H_{\theta_i} x$	list mode with time-of-flight
Magnetic resonance imaging (MRI)	radio frequency	$y = Fx$	uniform or non-uniform sampling in k space
Cardiac MRI (parallel, non-uniform)	radio frequency	$y_{t,i} = F_t W_i x$ W_i : coil sensitivity	gated or not, retrospective registration
Optical diffraction tomography	coherent light	$y_i = W_i F_i x$	with holography or grating interferometry

Discretization

- Discretization: How to turn the problem into linear algebra
- Examples
 - Diffraction-limited convolution (Fluorescence microscopy)
 - MRI
 - CT

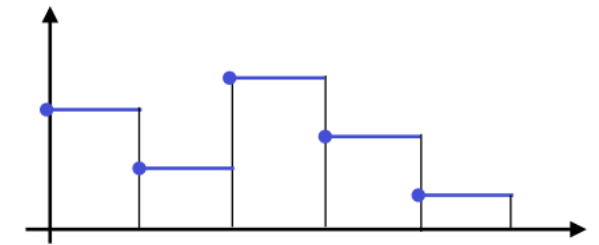
$$y = \mathbf{H}x + n$$



Discretization: Finite-Dimensional Formulation

Selection of an appropriate basis of functions $\beta_{\mathbf{k}} : \mathbb{R}^2 \rightarrow \mathbb{R}$ with $\mathbf{k} \in \Omega \subset \mathbb{Z}^2$ and $|\Omega| = N$.

Assume, *a priori*, that $s(\mathbf{r}) = \sum_{\mathbf{k} \in \Omega} s[\mathbf{k}] \beta_{\mathbf{k}}(\mathbf{r})$



Create a vector $\mathbf{x} = (s[\mathbf{k}])_{\mathbf{k} \in \Omega} \in \mathbb{R}^N$

- Measurement model (image formation):

$$y_m = \int_{\mathbb{R}^2} s(\mathbf{r}) \eta_m(\mathbf{r}) d\mathbf{r} + n[m] = \langle s, \eta_m \rangle + n[m], \quad m = 1, \dots, M$$

η_m : m th detector (analysis function)

$n[m]$: additive noise

$$[\mathbf{H}]_{m,\mathbf{k}} = \langle \eta_m, \beta_{\mathbf{k}} \rangle = \int_{\mathbb{R}^2} \eta_m(\mathbf{r}) \beta_{\mathbf{k}}(\mathbf{r}) d\mathbf{r}$$

Reshape into $\mathbf{H} \in \mathbb{R}^{M \times N}$:
 $\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$

We have now reduced the problem to a linear algebra problem
(given $\mathbf{x}$, we can synthesize $s(\mathbf{r})$)

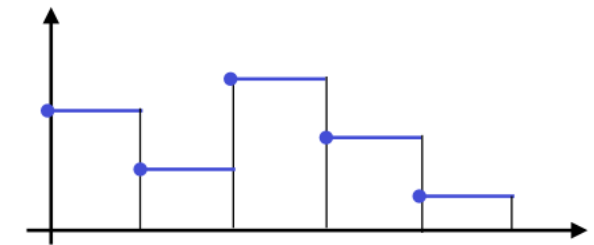
Examples of Basis Functions

Shift-invariant representation: $\beta_{\mathbf{k}}(\mathbf{x}) = \beta(\mathbf{x} - \mathbf{k})$

Separable generator: $\beta(\mathbf{x}) = \beta(x)\beta(y)$

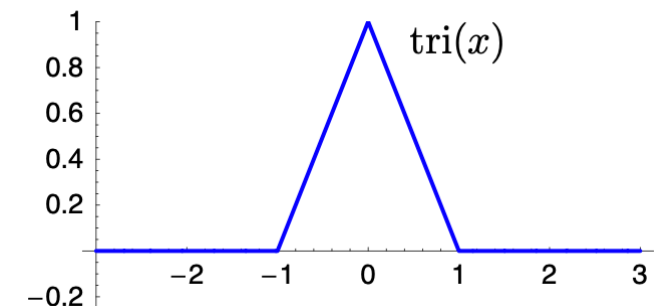
- Pixel basis:

$$\beta(x) = \text{rect}(x)$$



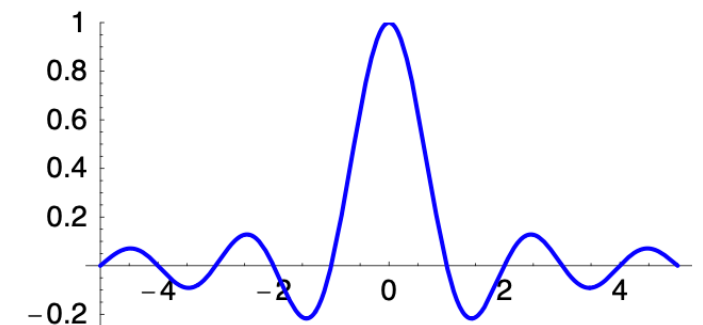
- Bilinear basis:

$$\beta(x) = (\text{rect} * \text{rect})(x) = \text{tri}(x)$$



- Bandlimited basis:

$$\beta(x) = \text{sinc}(x)$$



Example 1: Diffraction-Limited Convolution

Hypothesis: $\hat{h}_{2D}(\boldsymbol{\omega}) = 0$ for $\|\boldsymbol{\omega}\| \geq \omega_0$ (diffraction-limited)



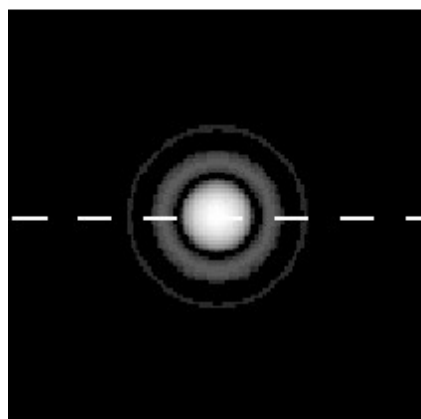
- Discretization

$\omega_0 \leq \pi$ and representation in (separable) sinc basis $\beta_{\mathbf{k}}(\mathbf{x}) = \text{sinc}(\mathbf{x} - \mathbf{k})$

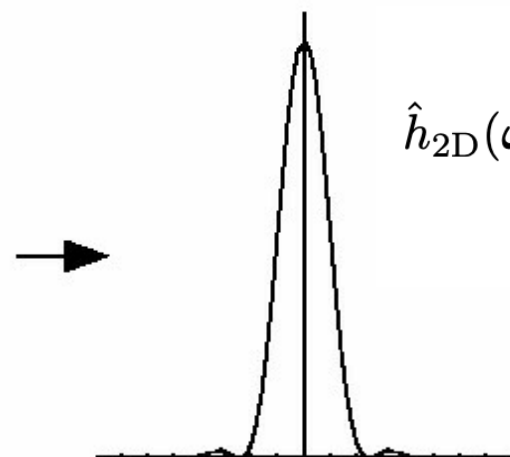
Analysis functions: $\eta_{\mathbf{m}}(\mathbf{x}) = h_{2D}(\mathbf{x} - \mathbf{m})$

$$[\mathbf{H}]_{\mathbf{m},\mathbf{k}} = \langle \eta_{\mathbf{m}}(\mathbf{x}), \text{sinc}(\mathbf{x} - \mathbf{k}) \rangle$$

$$= h_{2D}(\mathbf{m} - \mathbf{k})$$



Airy Disk



Radial Profile

$$\hat{h}_{2D}(\boldsymbol{\omega}) = \begin{cases} \frac{2}{\pi} \left(\arccos\left(\frac{\|\boldsymbol{\omega}\|}{\omega_0}\right) - \frac{\|\boldsymbol{\omega}\|}{\omega_0} \sqrt{1 - \left(\frac{\|\boldsymbol{\omega}\|}{\omega_0}\right)^2} \right), & \text{for } 0 \leq \|\boldsymbol{\omega}\| < \omega_0 \\ 0, & \text{otherwise} \end{cases}$$

Example 2: Magnetic Resonance Imaging (MRI)

- Simplified forward model for MRI

$$\hat{s}(\boldsymbol{\omega}_m) = \int_{\mathbb{R}^2} s(\boldsymbol{x}) e^{-j\boldsymbol{\omega}_m^T \boldsymbol{x}} d\boldsymbol{x}$$

(sampling of the Fourier transform)

$$\eta_m(\boldsymbol{x}) = e^{j\boldsymbol{\omega}_m^T \boldsymbol{x}}$$

suppose that $\|\boldsymbol{\omega}_m\|_2 < \pi$

- Discretization in separable sinc basis

$$\begin{aligned} [\mathbf{H}]_{m,\boldsymbol{k}} &= \langle \eta_m(\boldsymbol{x}), \text{sinc}(\boldsymbol{x} - \boldsymbol{k}) \rangle \\ &= \langle e^{j\boldsymbol{\omega}_m^T \boldsymbol{x}}, \text{sinc}(\boldsymbol{x} - \boldsymbol{k}) \rangle = e^{-j\boldsymbol{\omega}_m^T \boldsymbol{k}} \end{aligned}$$

Example 3: Computed Tomography (CT)

- Radon transform (line integrals)

Notation: $\boldsymbol{\theta} = (\cos \theta, \sin \theta) \in \mathbb{R}^2$

$$\mathcal{R}_{\boldsymbol{\theta}}\{f\}(t) = \int_{\mathbb{R}^2} f(\mathbf{x}) \delta(\boldsymbol{\theta}^\top \mathbf{x} - t) \, d\mathbf{x}$$

Recall the Radon transform of a separable function $\varphi(\mathbf{x}) = \varphi_1(x)\varphi_2(y)$:

$$\mathcal{R}_{\boldsymbol{\theta}}\{\varphi\}(t) = \varphi_{\boldsymbol{\theta}}(t),$$

where

$$\varphi_{\boldsymbol{\theta}}(t) = \frac{1}{|\cos \theta|} \varphi_1\left(\frac{t}{\cos \theta}\right) * \frac{1}{|\sin \theta|} \varphi_2\left(\frac{t}{\sin \theta}\right)$$

$$[\mathbf{H}]_{(i,j),\mathbf{n}} = \mathcal{R}_{\boldsymbol{\theta}_i}\{\varphi(\mathbf{x} - \mathbf{n})\}(t_j) = \varphi_{\boldsymbol{\theta}_i}(t_j - \mathbf{n}^\top \boldsymbol{\theta}_i)$$

Finite-Dimensional Inverse Problem

- Discretized forward model:

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

$$\mathbf{x} \in \mathbb{R}^N$$

$$\mathbf{H} \in \mathbb{R}^{M \times N}$$

$$\mathbf{y} \in \mathbb{R}^M$$

$$\mathbf{n} \in \mathbb{R}^M$$

- Inverse problem: How to efficiently recover $\mathbf{x}$ from $\mathbf{y}$?